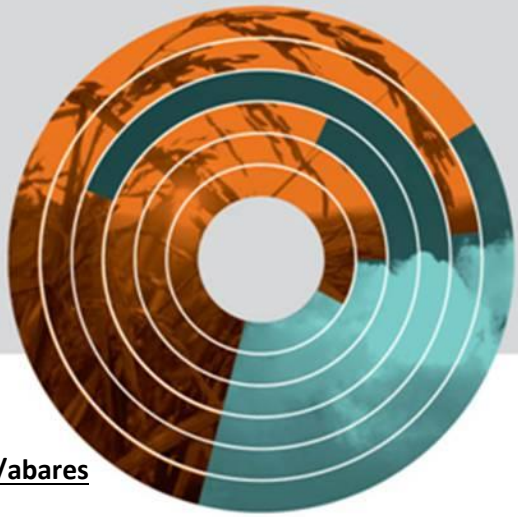
Australian Government
Department of Agriculture and Water Resources
ABARES

Weekly Australian Climate, Water and Agricultural Update



13 October 2016

The full report is available from **http://www.agriculture.gov.au/abares**

Summary of key issues

- During the week ending 12 October 2016, rainfall was recorded across southern Australia, parts of northern Australia and much of Tasmania.
- Maximum temperatures were generally average across most of Australia for the week ending 11 October 2016. Much of north-eastern Australia and southern Western Australia recorded minimum temperatures below average for this time of year.
- Special Climate Statement 58 – *Record September rains continue wet period in much of Australia*, released by the Bureau of Meteorology on 12 October 2016, reports that September 2016 was an exceptionally wet month over most of the eastern two-thirds of mainland Australia, as a succession of rain-bearing systems affected various parts of the continent. Averaged over Australia as a whole, September 2016 was the second-wettest September on record, just behind September 2010. It was the wettest September on record for New South Wales and the Northern Territory, as well as for the Murray–Darling Basin.
- The Bureau of Meteorology's Australian Tropical Cyclone Outlook indicates that an average to above-average number of cyclones are expected for the 2016–17 Australian tropical cyclone season (November–April).
- According to the Australian Plague Locust Commission, the outlook for the remainder of spring is for localised low and medium density locust nymphs in all regions of New South Wales; parts of the Far North, Western Agricultural and Northeast regions of South Australia; and in Northwest and North Central Victoria.
- The forecast for the next eight days indicates that rainfall is expected across northern, eastern and southern areas of Australia. No rainfall is forecast for most of western and central Australia.
- Water storage levels in the Murray–Darling Basin (MDB) increased by 167 gigalitres (GL) during the week ending 13 October 2016 to 18 111 GL, and are at 80 per cent of total capacity—the highest in over 35 months.
- In New South Wales, general security allocations increased in the Border Rivers (B class), Lower Namoi and Gwydir systems.
- Australian cattle prices continue to be supported by tight supplies with 2016–17 year-to-date (week ending 8 October 2016) cattle yardings down 26 per cent from the same period last year. Tight cattle supplies have also been reflected in reduced cattle slaughter, with throughput for the week ending 7 October 2016 down 19 per cent from that recorded over the same week last year.
- Market prices for mutton increased in the week ending 7 October 2016 with wet weather particularly affecting yardings. Graziers also continue to retain more sheep for flock rebuilding in response to the improved seasonal conditions. The resulting increased competition for available sheep between graziers and meat processors has driven mutton prices higher during 2016.

1. Climate

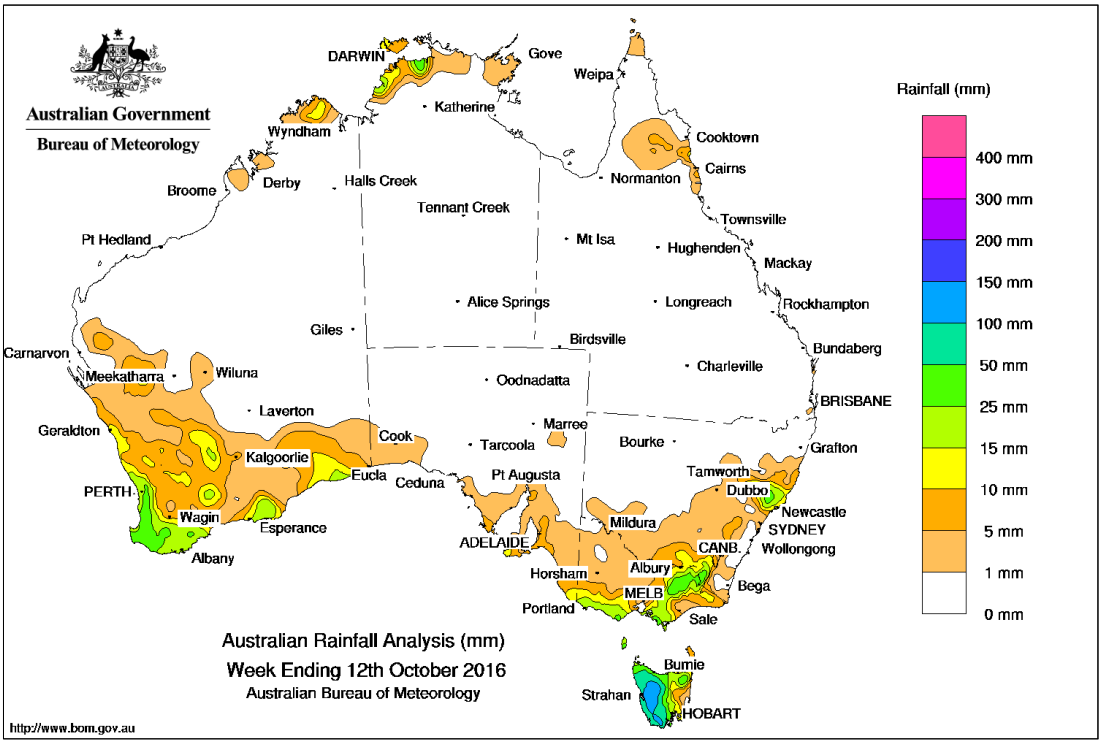
1.1. Rainfall this week

For the week ending 12 October 2016, rainfall was recorded across southern Australia, parts of northern Australia and much of Tasmania. Rainfall totals exceeding 15 millimetres were recorded across parts of south-eastern and the Central Coast of New South Wales, eastern and southern coast of Victoria, the far south-west of Western Australia and far north of the Northern Territory. Rainfall totals exceeding 100 millimetres were recorded in western Tasmania where the highest recorded weekly total was 158 millimetres at Mount Read.

The rainfall analyses and associated maps utilise data contained in the Bureau of Meteorology climate database, the Australian Data Archive for Meteorology (ADAM). The analyses are initially produced automatically from real-time data with limited quality control. They are intended to provide a general overview of rainfall across Australia as quickly as possible after the observations are received.

For further information go to http://www.bom.gov.au/climate/rainfall/

Rainfall for the week ending 12 October 2016



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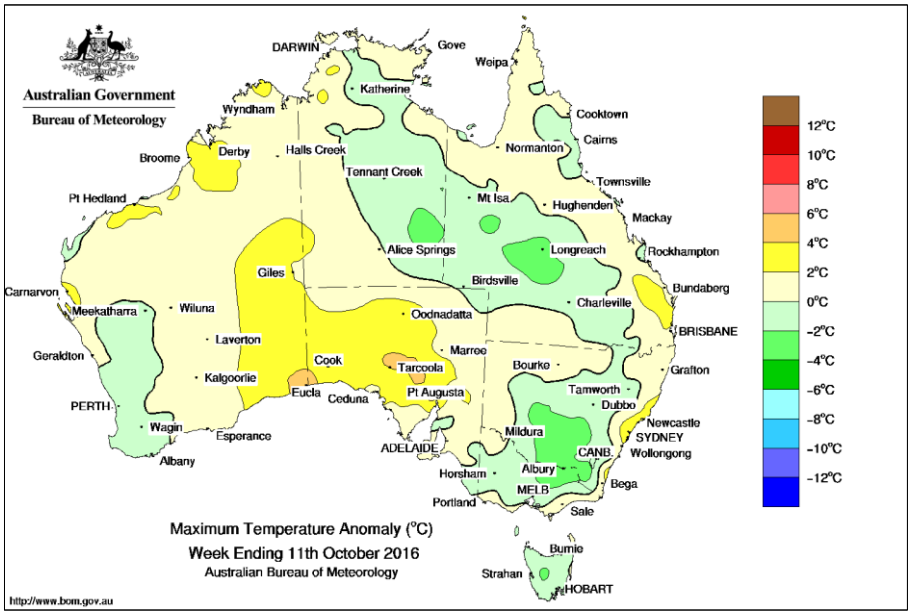
Issued: 12/10/2016

1.2. Temperature anomalies this week

For the week ending 11 October 2016, maximum temperatures were generally average across most of Australia. Parts of eastern Australia recorded maximum temperatures between 2°C and 4°C below average for this time of year. In contrast, some areas of southern Australia recorded minimum temperatures between 4°C and 6°C above average for this time of year.

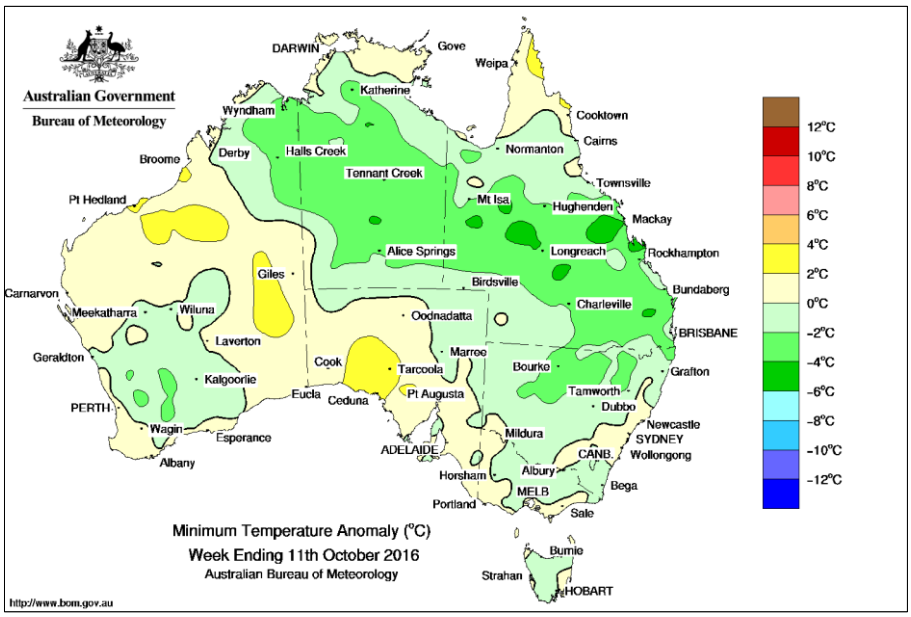
Minimum temperatures were below average across much of north-eastern Australia and southern Western Australia for the week ending 11 October 2016. Areas in central Queensland and the Northern Territory recorded below average minimum temperatures of between 4°C and 6°C for this time of year. Meanwhile, the remainder of the Australia recorded minimum temperatures closer to average or slightly above average for this time of year.

Maximum temperature anomalies for the week ending 11 October 2016



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Minimum temperature anomalies for the week ending 11 October 2016



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Note: Spatial temperature analyses are based on historical weekly temperature data provided by the Bureau of Meteorology. These temperature anomaly maps show the departure of the maximum and minimum temperatures from their long-term averages. Temperature anomalies are calculated using high-resolution gridded datasets from 1911 onwards. For further information go to http://www.bom.gov.au/jsp/awap/temp/index.jsp

1.3. Special Climate Statement

Special Climate Statement 58 – *Record September rains continue wet period in much of Australia*, released by the Bureau of Meteorology on 12 October 2016, reports that September 2016 was an exceptionally wet month over most of the eastern two-thirds of mainland Australia, as a succession of rain-bearing systems affected various parts of the continent. Monthly rainfall was at least double the long-term average over almost all of inland New South Wales and Queensland, most of the Northern Territory and outback South Australia, and parts of northern and western Victoria and eastern South Australia.

Averaged over Australia as a whole, September 2016 was the second-wettest September on record, just behind September 2010. It was the wettest September on record for New South Wales and the Northern Territory, as well as for the Murray–Darling Basin, while it ranks second-wettest for Victoria, third-wettest for Queensland and fourth-wettest for South Australia. For further information refer to the table below.

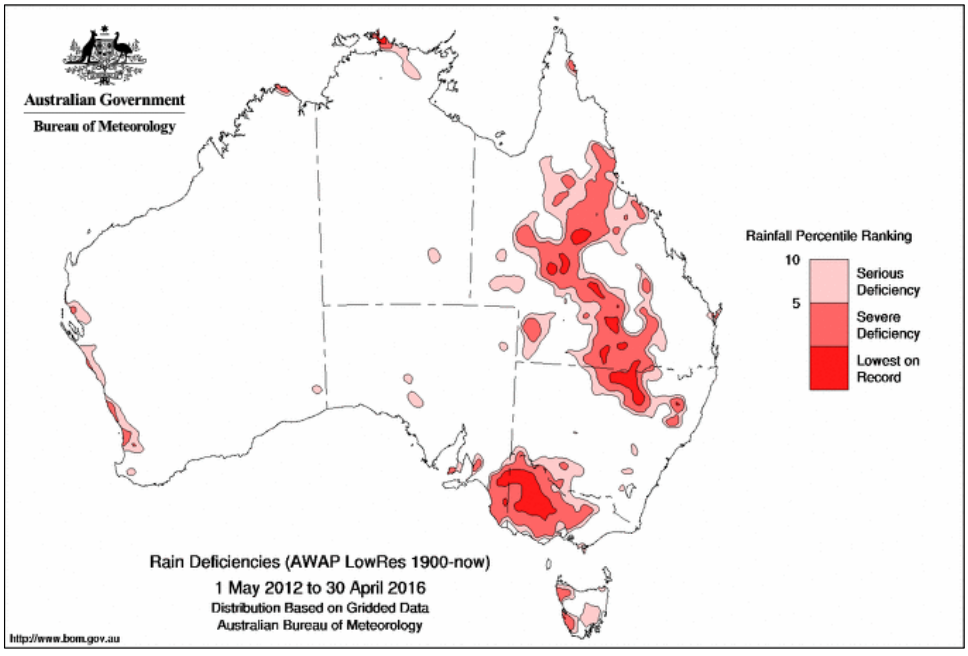
Area-averaged rainfall for September 2016.

Region	Rainfall (mm)	% above/ below average	Rank	Previous record
Australia	49.15	+196	2	50.83 (2010)
Queensland	60.87	+372	3	80.63 (2010)
New South Wales	113.69	+227	1	90.26 (1903)
Victoria	129.27	+98	2	134.38 (1916)
Tasmania	143.99	+4	45	256.69 (1928)
South Australia	46.94	+174	4	54.09 (2010)
Western Australia	7.82	−26	72	35.55 (1904)
Northern Territory	55.22	+688	1	41.65 (2010)
Murray–Darling Basin	121.39	+257	1	91.69 (1906)

The heavy rainfall during September continued a sequence of wet months which began in May with the breakdown of the 2015–16 El Niño. The May to September 2016 period was Australia's wettest on record, with each of the five individual months ranking in the 10 wettest in the past 117 years. Rainfall over this period was above average over almost all of Australia, except for parts of southwest Western Australia.

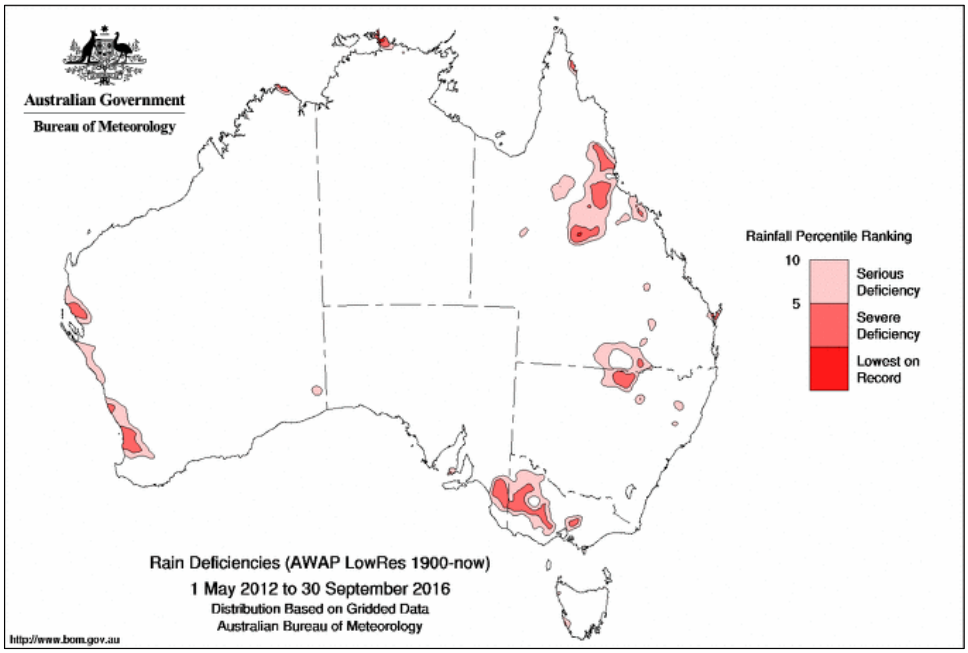
Prior to May 2016, many parts of eastern Australia were experiencing significant drought. Long-term drought since early 2012 had affected much of inland Queensland and adjacent border areas of northern New South Wales, as well as the western half of Victoria and southeast South Australia, while medium-term drought since mid- to late 2014 had been especially significant in Victoria (except Gippsland), Tasmania and eastern South Australia. The rains since May have eliminated most of the areas of medium-term rainfall deficiencies. They have also largely eliminated the long-term deficiencies in Queensland, except for some northern areas, and have significantly eased long-term deficiencies in western Victoria, as can be seen in the two maps below.

Australian rainfall deficiencies for the 48-month period May 2012 to April 2016



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Australian rainfall deficiencies for the period 53-month period May 2012 to September 2016



©Commonwealth of Australia 2016, Australian Bureau of Meteorology Issued: 05/10/2016

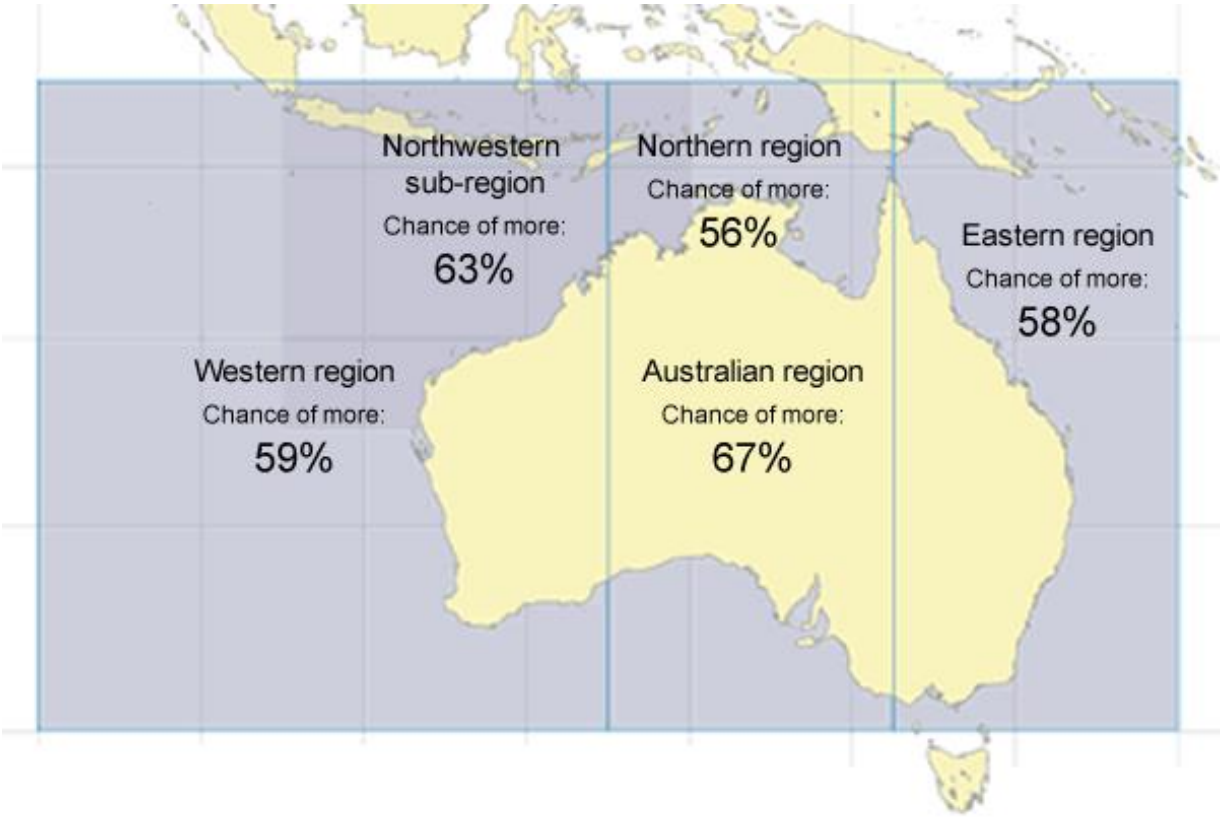
For further information go to: http://www.bom.gov.au/climate/current/statements/

1.4. Australian Tropical Cyclone Outlook

According to the Bureau of Meteorology, an average to above-average number of cyclones are expected for the 2016–17 Australian tropical cyclone season (November–April). Neutral to weak La Niña conditions in the tropical Pacific Ocean and warmer than average ocean temperatures to the north and east of Australia have influenced this year's tropical cyclone outlook. During neutral years, the first tropical cyclone to make landfall typically occurs in late December. In La Niña years, the first cyclone to make landfall over Australia typically occurs earlier, around the first week of December. The Australian region typically experiences more tropical cyclone activity during La Niña years.

This outlook is based upon the status of the El Niño-Southern Oscillation (ENSO) over the preceding July to September period. Sea surface temperatures in the tropical Pacific Ocean are showing some La Niña-like characteristics but have not reached La Niña thresholds. The majority of climate models predict these ocean conditions to continue into summer with only one model predicting La Niña thresholds to be exceeded. Ocean temperatures are currently 1–2 °C warmer than average to the north and east of Australia which is favourable for tropical cyclone development. The Southern Oscillation Index, a measure of the atmospheric component of ENSO, has been positive through this period and exceeded La Niña thresholds in the last two weeks of September.

Chance of more tropical cyclones than average for 2016 to 2017



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1.5. Australian Plague Locust

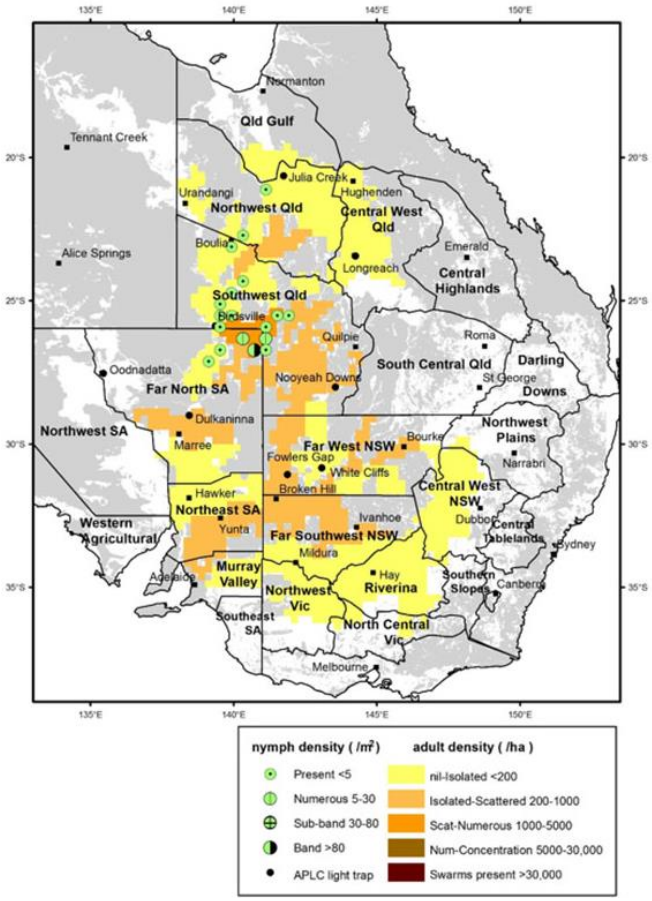
The Australian Plague Locust Commission (APLC) produces a monthly Locust Bulletin during the seasons of locust activity (spring, summer and autumn). The bulletin gives regional summaries of the locust situation and weather events of potential significance for locust development. It also provides a forecast of likely developments for the next two months (or from autumn to spring) for the Australian plague locust *Chortoicetes terminifera*, spur-throated locust *Austracris guttulosa,* and the migratory locust *Locusta migratoria.*

According to the APLC, autumn locust distribution in most regions of New South Wales, Victoria, Queensland and South Australia was generally at low densities. However, localised high density nymphs and adults persisted in part of Southwest Queensland and the Far North region of South Australia during May. Breeding by that population produced a widespread low and medium density nymphal population in mid-August. Surveys identified nymphs at all development stages. The widespread heavy rainfall during winter and early spring throughout all regions of eastern Australia has produced unseasonal vegetation growth in arid and summer rainfall areas. It has also produced dense short-lived pasture growth in southern New South Wales, South Australia and Victoria. At this stage it is unclear if dense vegetation in southern regions will favour locust development, but it will provide opportunities for spring breeding in Queensland and northern regions of South Australia and New South Wales. Flooding and road closures have limited any APLC surveys and little information is available on current population or development stages.

The outlook for the remainder of spring is for localised low and medium density nymphs in all regions of New South Wales; parts of the Far North, Western Agricultural and Northeast regions of South Australia; and in Northwest and North Central Victoria. In Southwest Queensland and the Far North region of South Australia, adult numbers are likely to increase to medium densities. Breeding could occur in October and produce a late spring nymph generation, resulting in a significant adult population increase in November or December. There is a low risk of widespread high density infestations in any region during spring 2016.

For further information, go to http://www.agriculture.gov.au/pests-diseases-weeds/locusts/

Australian Plague Locust Distribution
1 August to 30 September 2016



Densities estimated for areas of locust habitat, based on current surveys and reports from current and previous month.

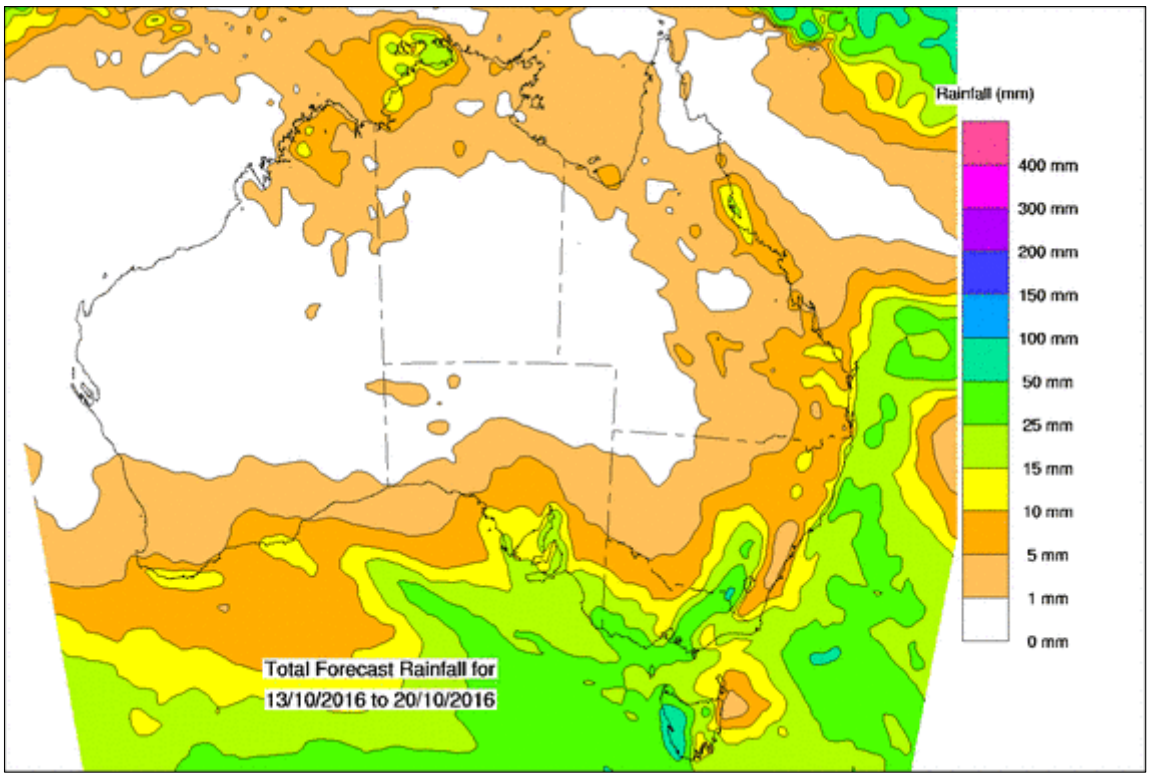
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1.6. Rainfall forecast

The forecast for the next eight days indicates that rainfall is expected across northern, eastern and southern areas of Australia. Rainfall totals exceeding 15 millimetres are forecast for an area in south-eastern New South Wales, eastern and southern Victoria, south-eastern South Australia and most of Tasmania. No rainfall is forecast across most of western and central Australia for the next eight days.

This rainfall forecast is produced from computer models. As it contains no input from weather forecasters, it is important to check local forecasts and warnings issued by the Bureau of Meteorology.

Total forecast rainfall (mm) for the period 13 to 20 October 2016



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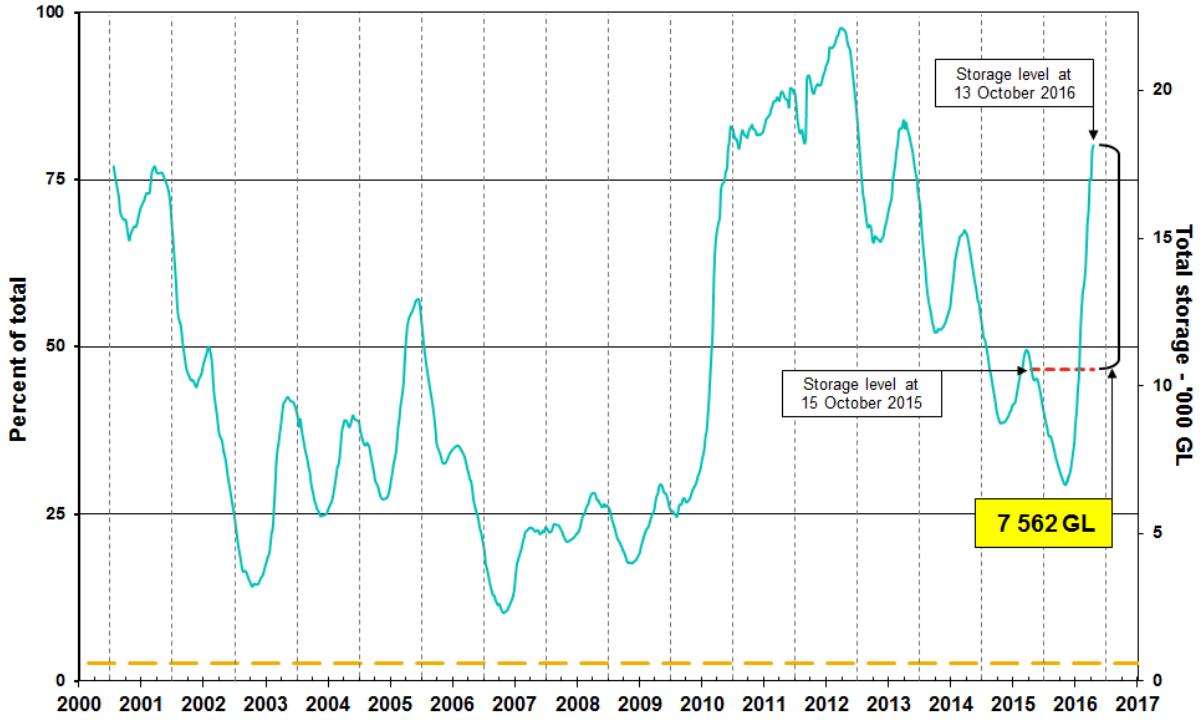
Issued: 13/10/2016

2. Water

2.1. Water availability

Water storage levels in the Murray–Darling Basin (MDB) increased by 167 gigalitres (GL) during the week ending 13 October 2016 to 18 111 GL, and are at 80 per cent of total capacity—the highest in over 35 months. This water storage level is 33 percentage points or 7 562 GL more than at the same time last year.

Water storages in the Murray–Darling Basin (NSW, Victoria and Queensland)



Information on irrigation water available in the Murray–Darling Basin from 1 January 2001 to 13 October 2016 is shown above. The top horizontal (short dash) line indicates the storage level during a similar time last year. The bottom horizontal (long dash) line indicates the amount of 'dead' or unusable storage.

2.2. Water allocations

The current water allocations for the 2016–17 water trading season for licence holders in the New South Wales, Victoria and South Australia water systems are summarised in the following table.

- In New South Wales, general security allocations increased in the Border Rivers (B class), Lower Namoi and Gwydir systems.

Allocations at	**13 October 2016 (%)**		**6 October 2016 (%)**	
New South Wales	General security	High security	General security	High security
NSW Murray Valley *	53	97	53	97
NSW Murrumbidgee Valley *	61	95	61	95
NSW Lower Darling *	100	100	100	100
NSW Macquarie and Cudgegong Valley *	100	100	100	100
NSW Hunter Valley	100	100	100	100
NSW Lachlan Valley *	124	100	124	100
NSW Lower Namoi *	101.22	100	59	100
NSW Upper Namoi *	100	100	100	100
NSW Gwydir Valley *	61.44	100	32.51	100
NSW Border Rivers *	100 [a] / 73.1 [b]	100	100 [a] / 39.9 [b]	100
NSW Peel Valley	100	100	100	100
Victoria	Low reliability *	High reliability *	Low reliability *	High reliability *
Victoria Murray Valley	0	100	0	100
Victoria Goulburn	0	86	0	86
Victoria Campaspe	43	100	43	100
Victoria Loddon	0	86	0	86
Victoria Bullarook	100	100	100	100
Victoria Broken	52	100	52	100
South Australia		High security		High security
South Australia Murray Valley		100		100

*Carryover water may also be available

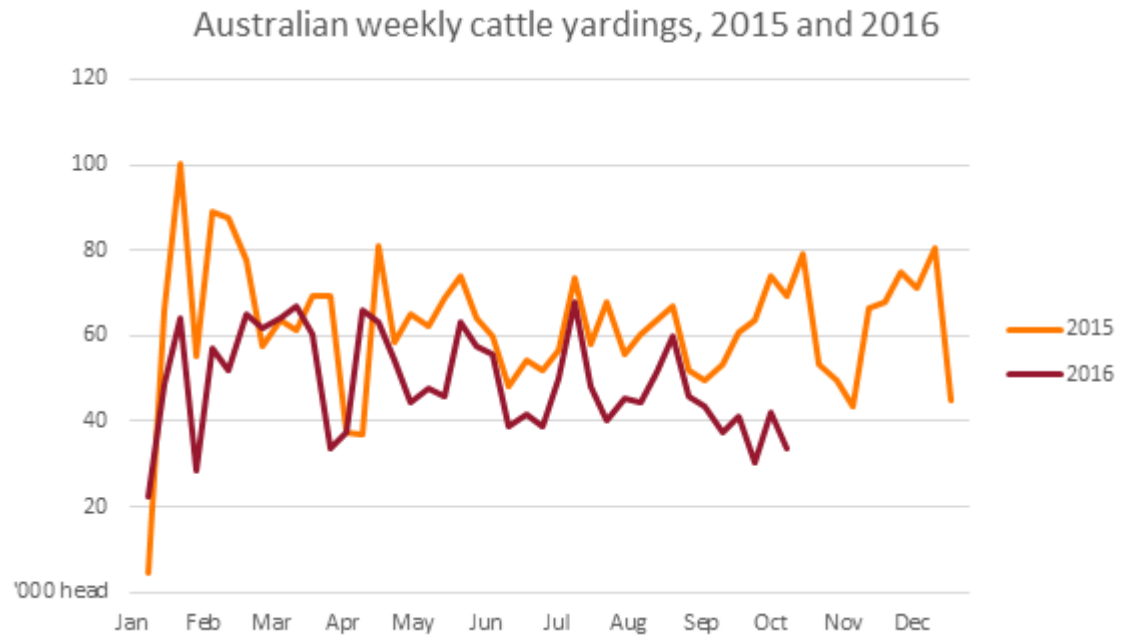
a General security A class. b General security B class

3. Commodities

3.1. Market focus

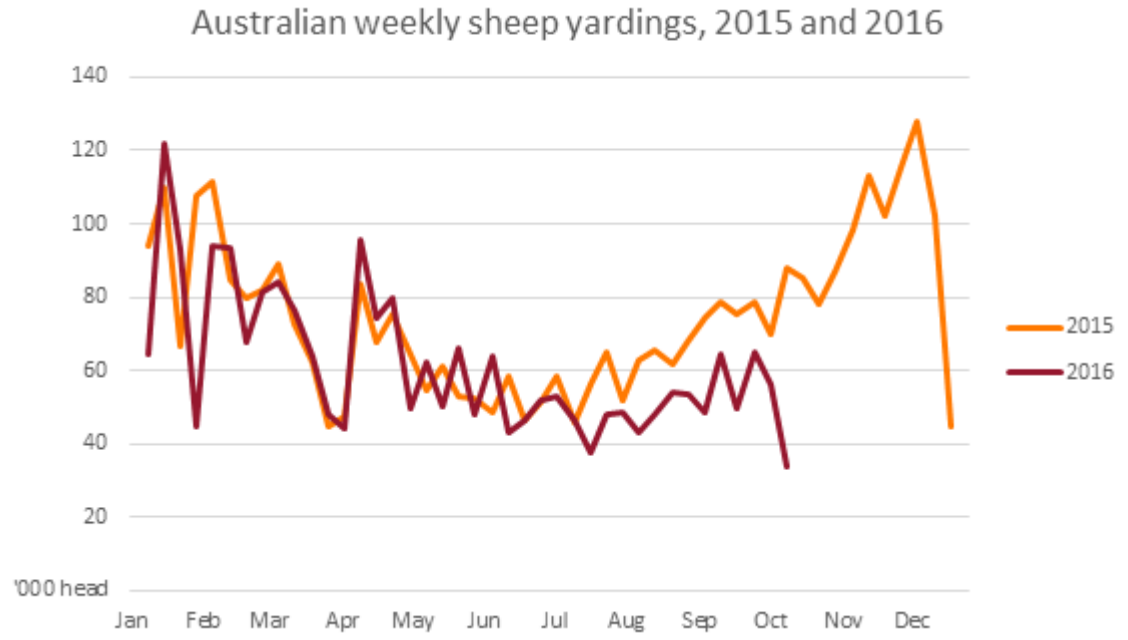
Beef

Australian cattle prices continue to be supported by tight supplies with 2016–17 year-to-date (week ending 8 October 2016) cattle yardings down 26 per cent from the same period last year. Tight cattle supplies have also been reflected in reduced cattle slaughter, with throughput for the week ending 7 October 2016 down 19 per cent from the same week last year.



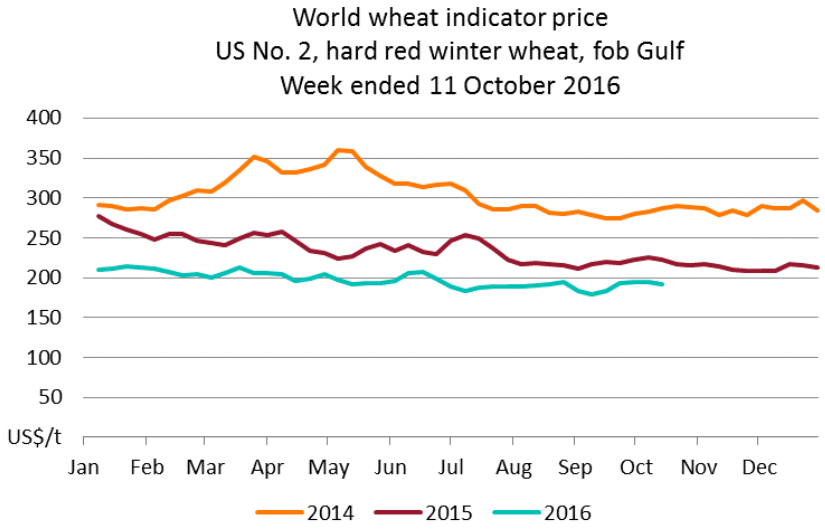
Mutton

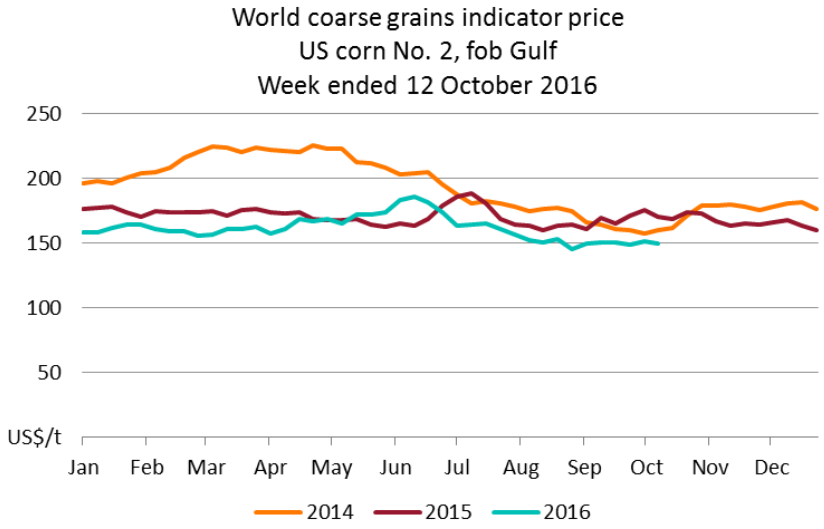
Market prices for mutton increased in the week ending 7 October 2016 with wet weather particularly affecting yardings. Graziers also continue to retain more sheep for flock rebuilding in response to the improved seasonal conditions. The resulting increased competition for available sheep between graziers and meat processors has driven mutton prices higher during 2016.

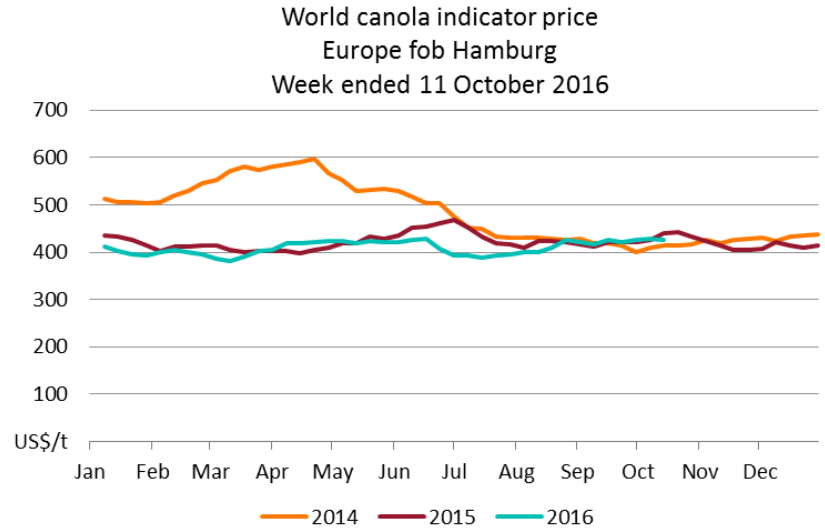


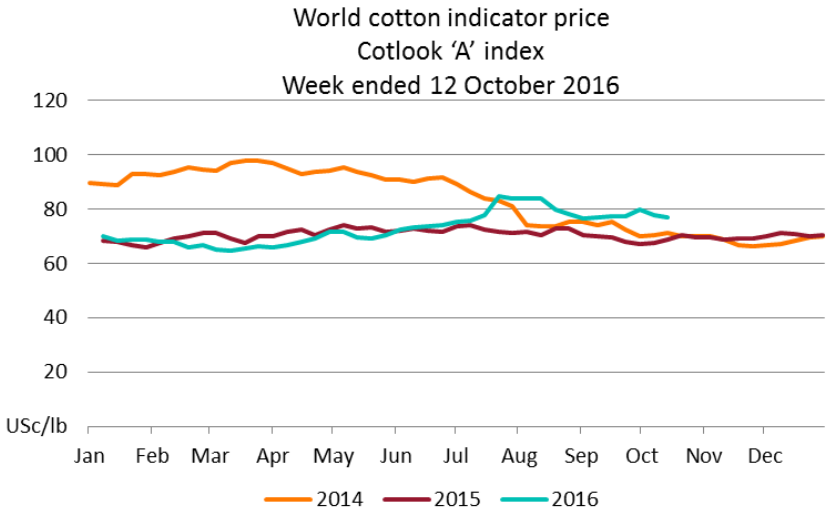
Indicator	Week ended	Unit	Latest Price	Price week prior	Weekly Change	Price 12 months prior	Year on year change
Selected World Indicator Prices							
Australian Dollar - AUD/USD Exchange Rate	12-Oct	US$/A$	0.76	0.77	-1% ⇩	0.73	4% ⇧
Wheat - US No. 2 hard red winter wheat, fob Gulf	11-Oct	US$/t	192	194	-1% ⇩	223	-14% ⇩
Coarse Grains - US No. 2 yellow corn, fob Gulf	12-Oct	US$/t	150	151	-1% ⇩	171	-12% ⇩
Canola - Rapeseed, Europe, fob Hamburg	11-Oct	US$/t	426	428	0% ⇩	439	-3% ⇩
Cotton - Cotlook 'A' Index	12-Oct	USc/lb	76.9	77.7	-1% ⇩	68.9	12% ⇧
Sugar - Intercontinental Exchange, nearby futures, no.11 contract	12-Oct	USc/lb	23.2	22.7	2% ⇧	14.1	65% ⇧
Wool - Eastern Market Indicator	06-Oct	Ac/kg clean	1 300	1 290	1% ⇧	1 195	9% ⇧
Global Dairy Trade (GDT) weighted average prices[a]							
Dairy - Whole milk powder	04-Oct	US$/t	2 681	2 782	-4% ⇩	2 824	-5% ⇩
Dairy - Skim milk powder	04-Oct	US$/t	2 209	2 293	-4% ⇩	2 267	-3% ⇩
Dairy - Cheddar cheese	04-Oct	US$/t	3 430	3 518	-3% ⇩	3 234	6% ⇧
Dairy - Anhydrous milk fat	04-Oct	US$/t	4 954	4 998	-1% ⇩	3 763	32% ⇧
Selected domestic crop indicator prices							
Grain Sorghum - Sydney, NSW	12-Oct	A$/t	214	204	4% ⇧	285	-25% ⇩
Feed Wheat - General purpose, Sydney, NSW	12-Oct	A$/t	246	241	2% ⇧	285	-14% ⇩
Feed Barley - Sydney, NSW	12-Oct	A$/t	187	184	2% ⇧	259	-28% ⇩
Canola - Portland, Vic.	10-Oct	A$/t	528	524	1% ⇧	536	-1% ⇩
Selected domestic livestock indicator prices							
Beef - Young cattle (300-400 kg live weight C3), Qld	30-Sep	Ac/kg cwt	653	684	-5% ⇩	557	17% ⇧
Mutton - Mutton indicator (18-24 kg fat score 2-3), Vic	07-Oct	Ac/kg cwt	458	420	9% ⇧	299	53% ⇧
Lamb - Lamb indicator (18-22 kg fat score 2-4), Vic	07-Oct	Ac/kg cwt	594	593	0% ⇧	483	23% ⇧
Pig - Pig indicator (Buyers 60.1-75 kg), NSW	30-Sep	Ac/kg cwt	373	373	0% ●	348	7% ⇧

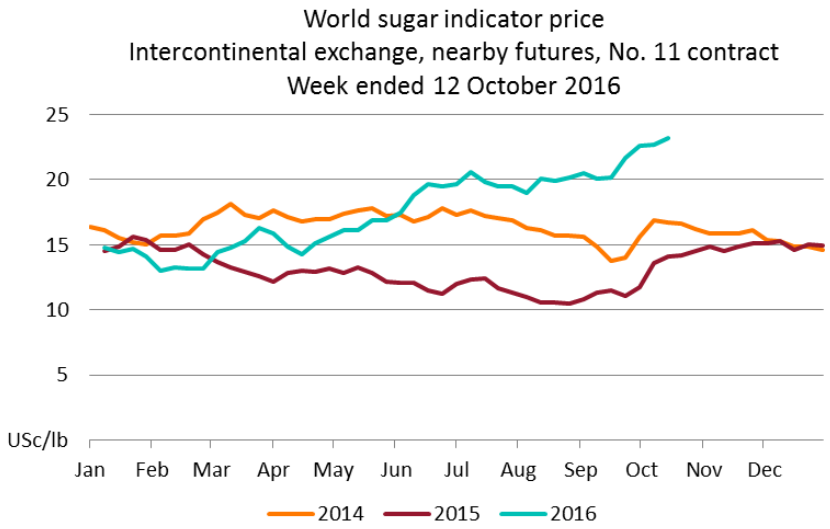
3.2. Selected world indicator prices

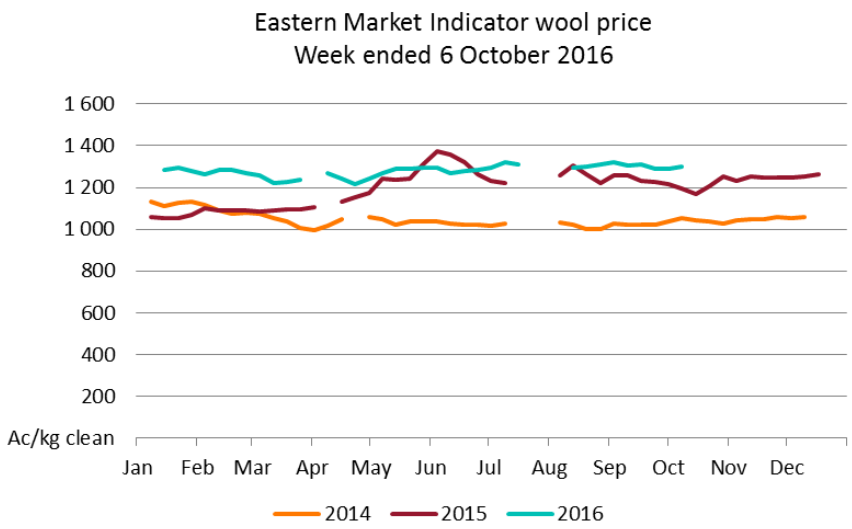


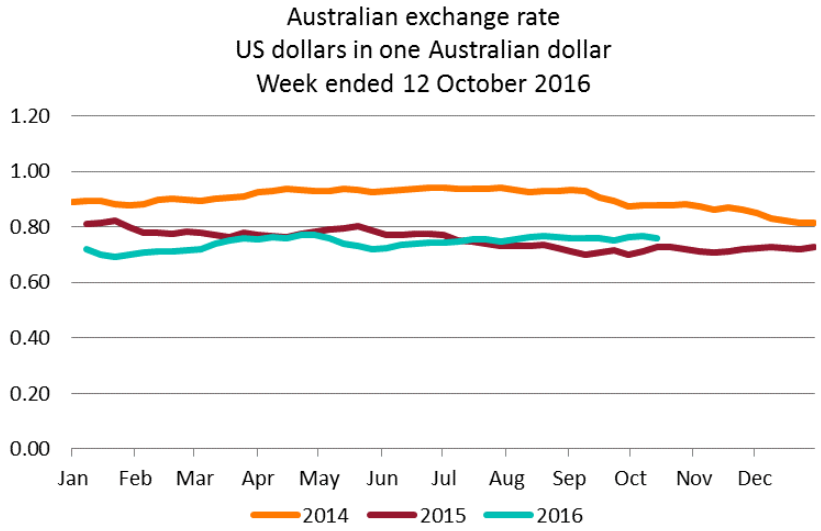




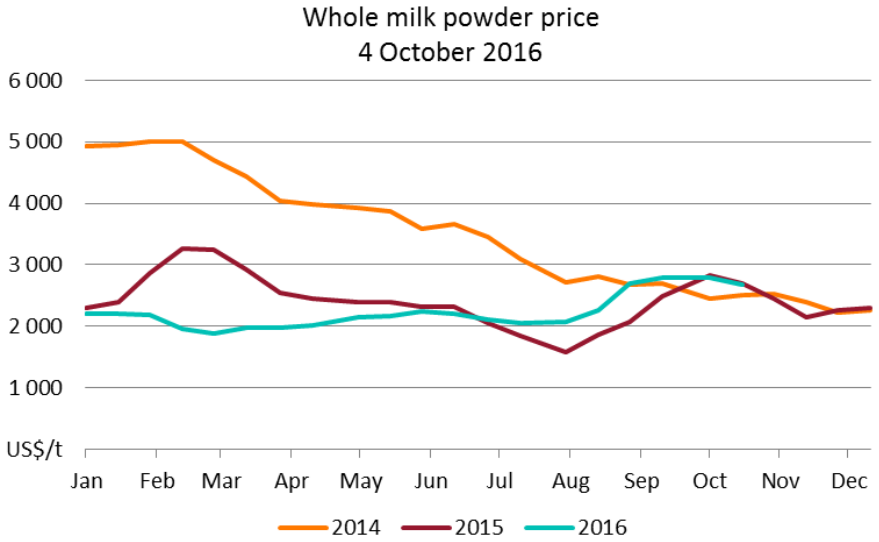


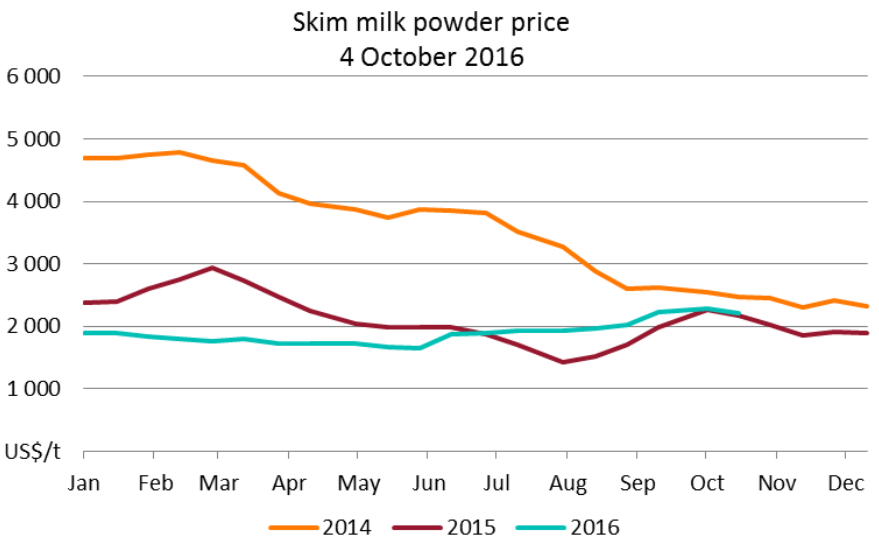

World sugar indicator price
Intercontinental exchange, nearby futures, No. 11 contract
Week ended 12 October 2016
25
20
15
10
5
USc/lb
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2014 2015 2016

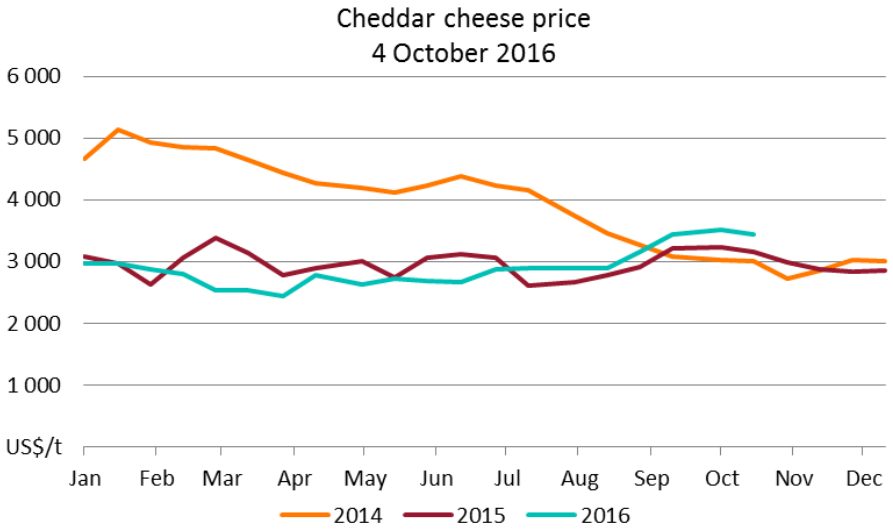

Eastern Market Indicator wool price
Week ended 6 October 2016
1 600
1 400
1 200
1 000
800
600
400
200
Ac/kg clean
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2014 2015 2016

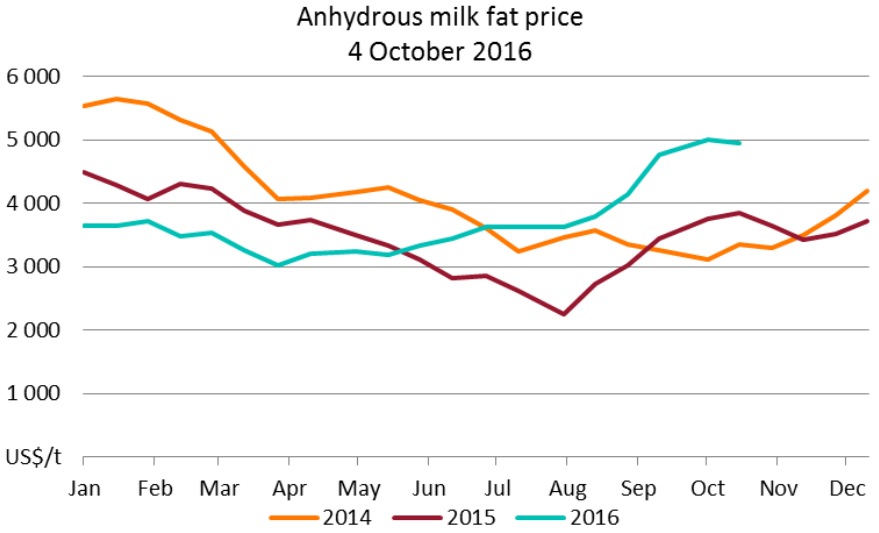

Australian exchange rate
US dollars in one Australian dollar
Week ended 12 October 2016
1.20
1.00
0.80
0.60
0.40
0.20
0.00
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2014 2015 2016

3.3. Global Dairy Trade (GDT) weighted average prices

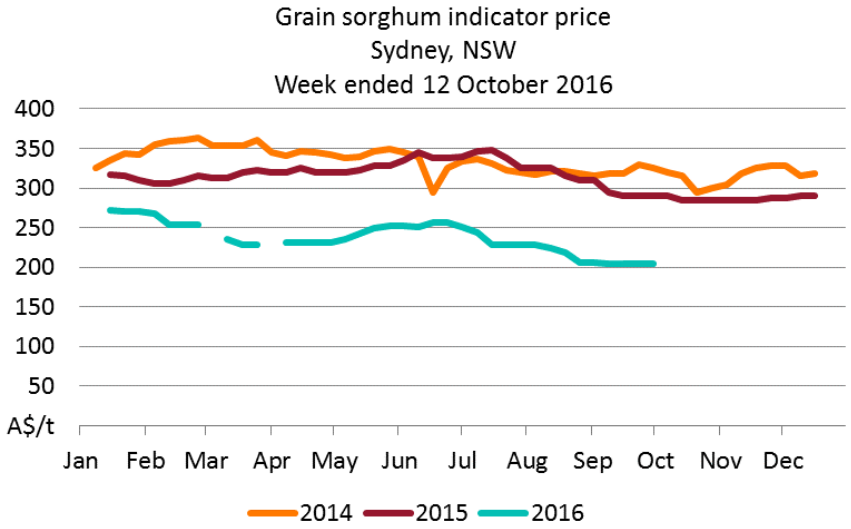


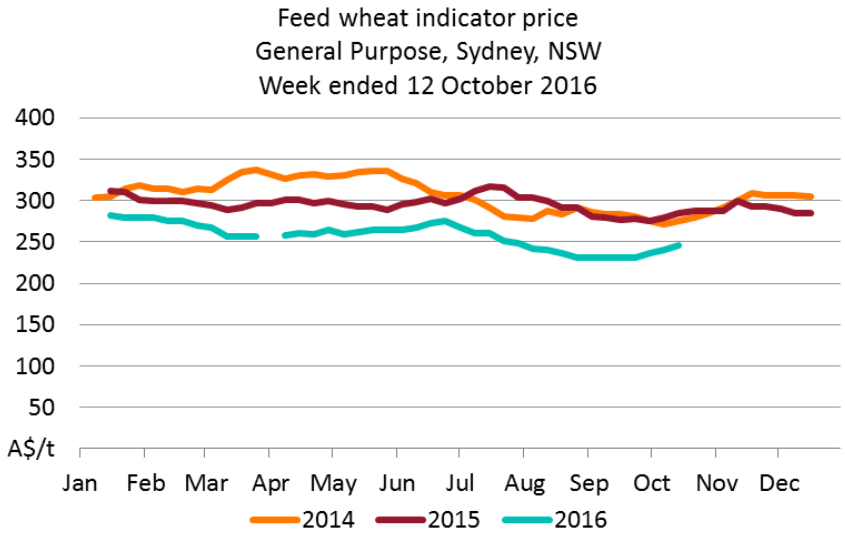


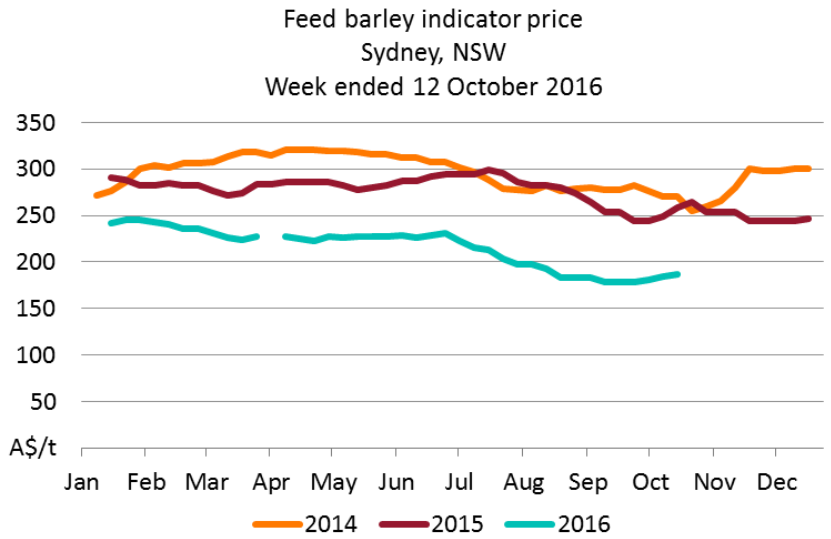


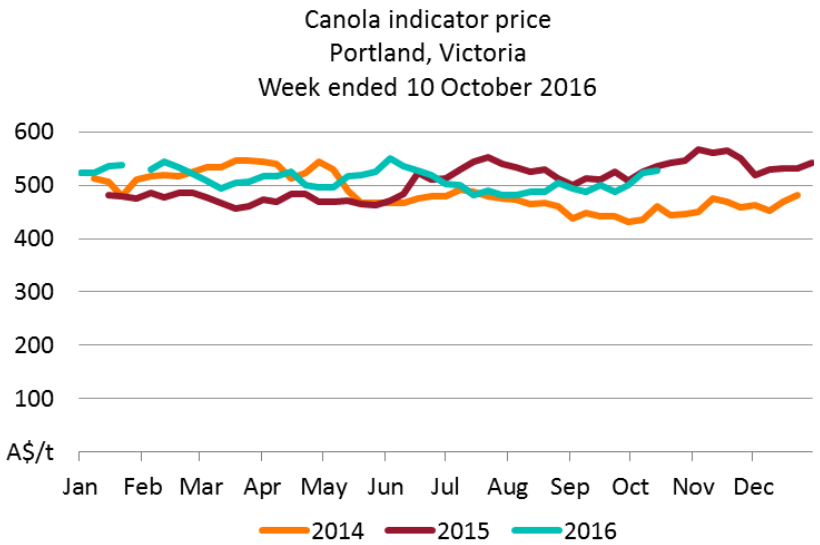


3.4. Selected domestic crop indicator prices

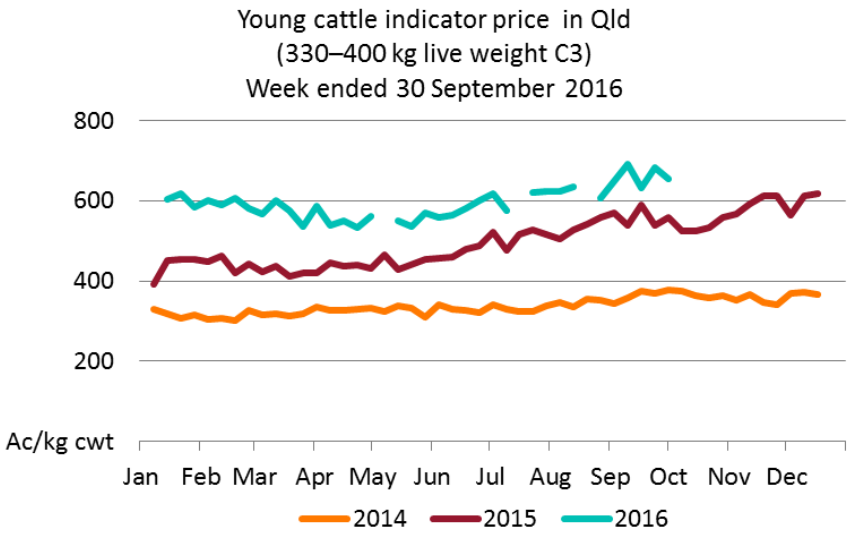


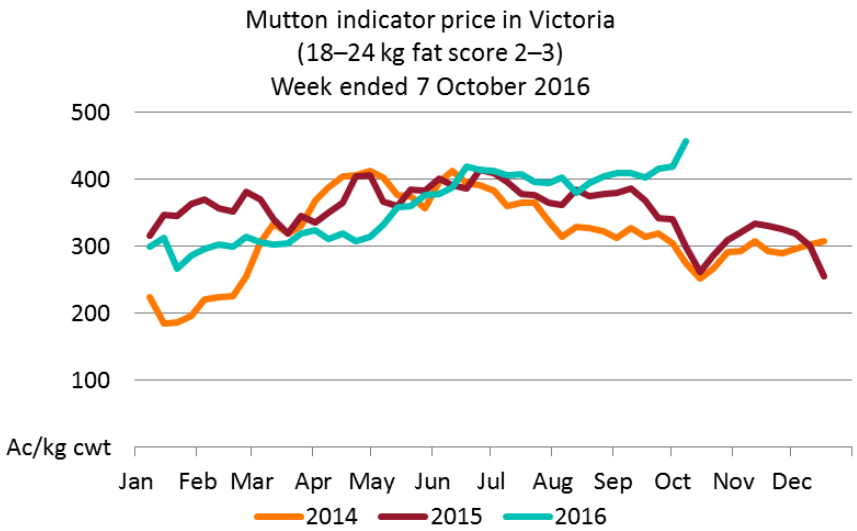


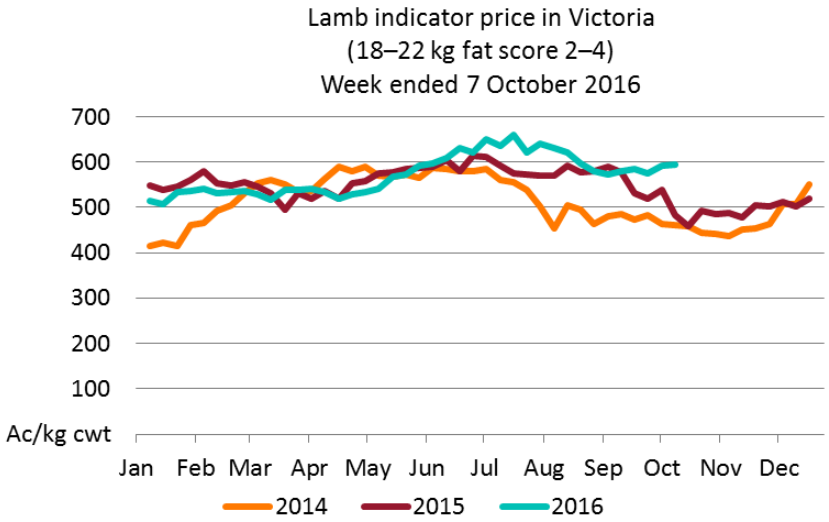


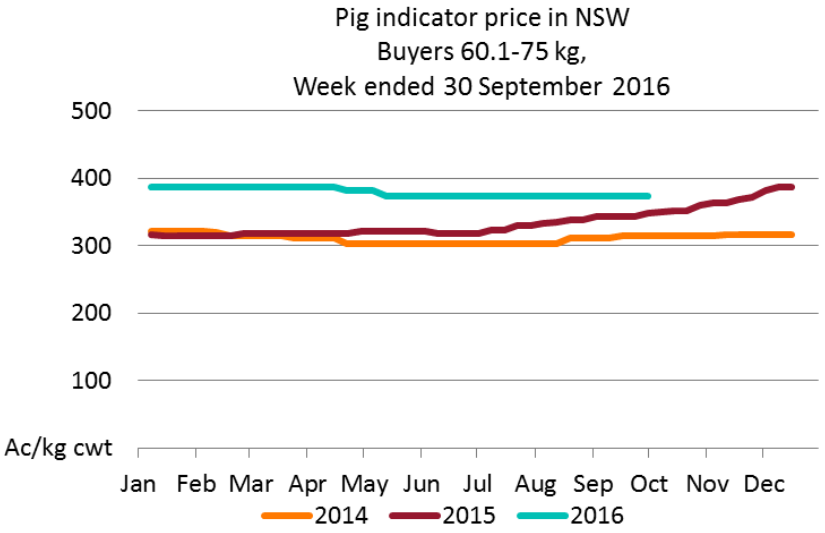


3.5. Selected domestic livestock indicator prices



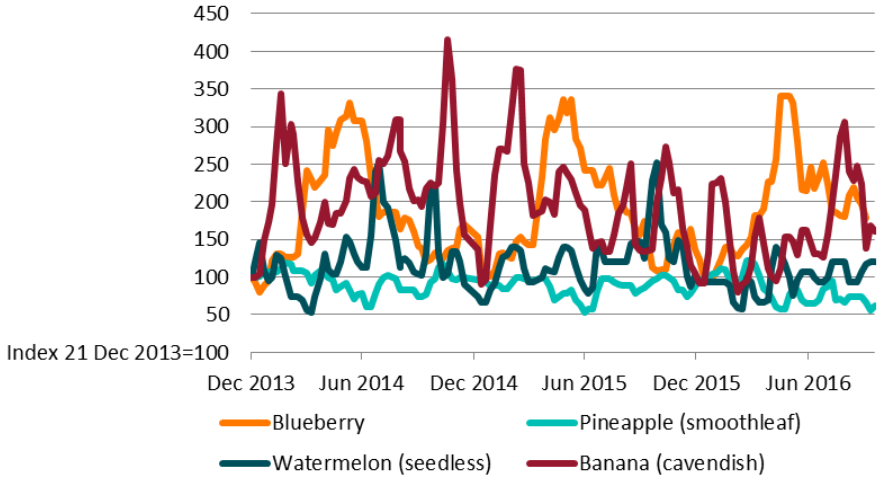




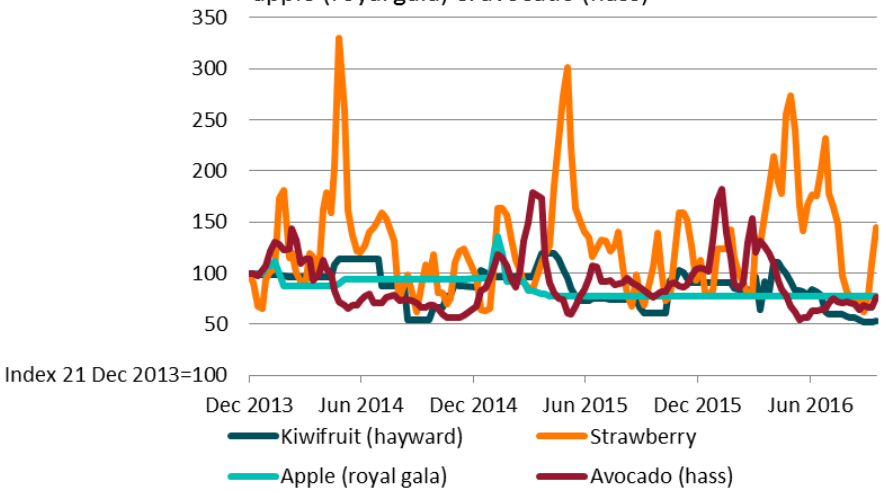


3.6. Movements in selected fruit and vegetable prices – Week ended 8 October 2016

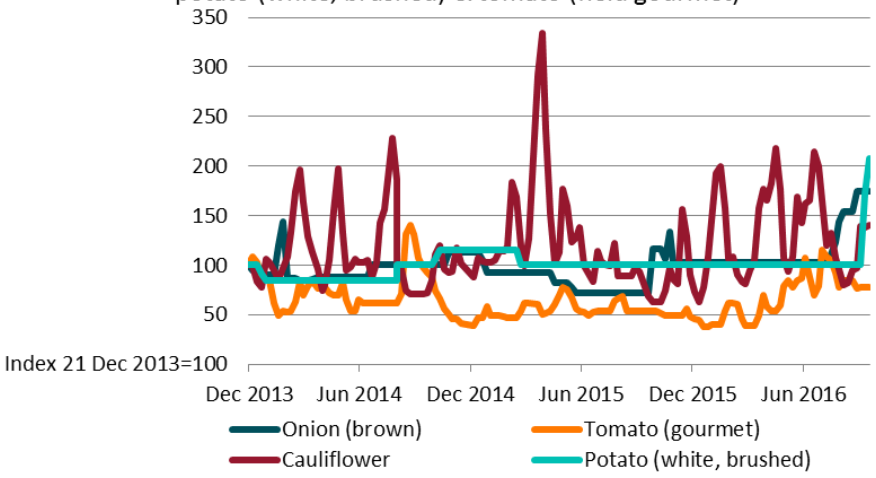
Weekly wholesale prices for blueberry, pineapple (smoothleaf), watermelon (seedless) & banana (cavendish)



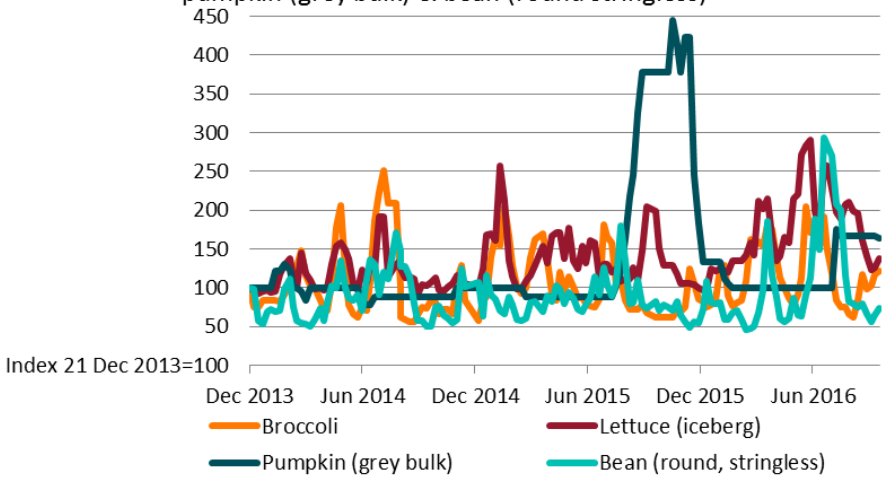
Weekly wholesale prices for kiwifruit (hayward), strawberry, apple (royal gala) & avocado (hass)



Weekly wholesale prices for onion (brown), cauliflower, potato (white, brushed) & tomato (field gourmet)



Weekly wholesale prices for broccoli, lettuce (iceberg), pumpkin (grey bulk) & bean (round stringless)



4. Data attribution

Climate

Bureau of Meteorology

- Weekly rainfall totals: www.bom.gov.au/jsp/awap/rain/index.jsp
- Monthly and last 3-months rainfall percentiles: www.bom.gov.au/jsp/awap/rain/index.jsp
- Temperature anomalies: www.bom.gov.au/jsp/awap/temp/index.jsp
- Rainfall forecast: www.bom.gov.au/jsp/watl/rainfall/pme.jsp
- Seasonal outlook: www.bom.gov.au/climate/ahead
- Drought statement: www.bom.gov.au/climate/drought/drought.shtml
- Soil moisture: Bureau of Meteorology (Australian Water Resources Assessment Landscape model) www.bom.gov.au/water/landscape/

Water

New South Wales

- New South Wales Water Information: http://waterinfo.nsw.gov.au/
- New South Wales Office of Water, Department of Primary Industries: www.water.nsw.gov.au/Home/default.aspx

Available water determinations register: http://www.water.nsw.gov.au/water-licensing/registers

Queensland

- Sunwater: www.sunwater.com.au
- Seqwater: http://seqwater.com.au

South Australia

- SA Water: https://www.sawater.com.au/community-and-environment/the-river-murray/river-reports/daily-flow-report
- South Australian Department of Environment, Water and Natural Resources: http://www.environment.sa.gov.au

Victoria

- Goulburn–Murray Water: www.g-mwater.com.au

Commodities

Fruit and vegetables

- Datafresh: http://www.freshstate.com.au

Mutton, lambs, wheat, barley and grain sorghum

- The Land: hardcopy or online at http://theland.farmonline.com.au/markets

Cattle, mutton and lambs

- Meat and Livestock Australia: http://www.mla.com.au/Prices-and-markets

Pigs

- Australian Pork Limited: www.australianpork.com.au

Canola

- Weekly Times: hardcopy

Dairy

- Global Dairy Trade: http://www.globaldairytrade.info/en/product-results/